Arul Mathur

arulster17@gmail.com | 408-750-4646

github.com/arulster17 | linkedin.com/in/arulster17

EDUCATION

University of California, San Diego — **June 2026**

B.S. Computer Science — *GPA: 3.99/4.0*

Courses: LLM Optimization, NLP, Parallel Computing, HPC, Computer Security, ML Algorithms, Quantum Cryptography, Computer Graphics, Operating Systems, Databases, Computer Architecture, Compilers, Search&Optimization

WORK EXPERIENCE

Research Assistant, University of California, Berkeley — **2021 - 2023**

Developed an AI-powered vulnerability detection framework for Android apps, combining fuzz testing with reinforcement learning to improve coverage and exploit discovery. Presented at Foothill Symposium.

Tutor, Jacobs School of Engineering, UCSD — **September 2025 - March 2026**

Tutor for CSE 101 (Design and Analysis of Algorithms) and CSE 127 (Computer Security).

Software Engineering and Curriculum Intern, Art of Problem Solving — **July - September 2024**

Built a simulator into AoPS IDE to enable UI-simulations of algorithms and automated testing for algorithmic correctness and complexity. Authored USA Computing Olympiad algorithmic problems and solutions.

Intern, Glider AI — **June - August 2023**

Created a Python-based marketing lead segmentation tool to enable enriched analysis of marketing funnels. Improved SEO performance with SEMRush, cluster pages, HubSpot, HARO, and other tools.

PROJECTS

- **SpecSplit LLM Serving:** Built a speculative decoding prototype for low-latency LLM inference across disaggregated hardware, implementing gRPC-based draft/verify orchestration, KV cache management, and asynchronous orchestration from scratch. Studied latency and throughput tradeoffs across heterogeneous hardware by analyzing acceptance rate, hit rate, and pipeline efficiency.
- **LLM Efficiency:** Optimized performance of Gemma-2b, an open-source text-to-text LLM, using sparse autoencoders and Logit Lens to ablate/steer redundant layers and reduce computations.
- **AI Personal Assistant**: Built a personal assistant bot using a fully AI-native architecture. Integrated Google Gemini for natural language intent routing across modular feature handlers (Calendar, YouTube, Search). Prioritized safety through least-privilege OAuth scoping and confirmation-gated destructive actions to protect user data.
- **Parallel Computing**: Optimized algorithms for matrix multiplication, convolution, and GeMM through parallelization with OpenCL. Built a raytracer and fractal rendering using CUDA.
- **Carden - Game Design:** Led a team of 10 to build a multi-level game. Authored design docs to align across backend, frontend, and design efforts. Implemented core backend logic and game mechanics.
- **Quantum Computing:** Implemented quantum algorithms (Shor's, Grover's, Simon's) with IBM's Qiskit and studied their cryptographic implications.
- **Operating Systems:** Implemented key operating systems capabilities like process synchronization, IPC, multithreading, weighted process scheduling, virtual memory management.
- **Network Programming:** Built a multithreaded client–server chat application in C using POSIX sockets and semaphores, enabling concurrent messaging across multiple clients on a local network.

CERTIFICATIONS

Machine Learning Specialization, Deep Learning Specialization

SKILLS

Frameworks: CUDA, OpenCL, PyTorch, NumPy, Android **Languages:** C, C++, Python, Java, JavaScript, SQL, Haskell